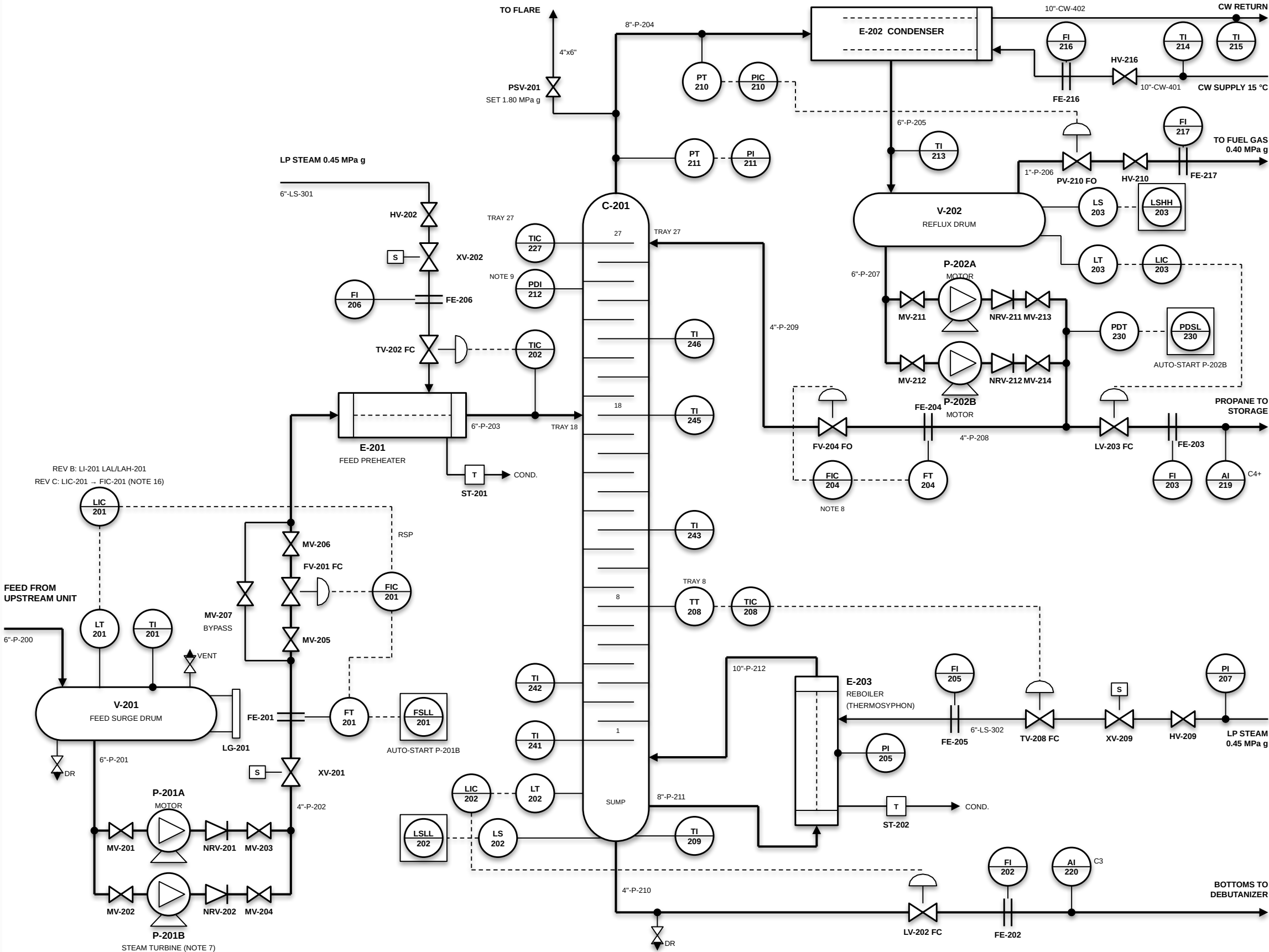




GENERAL NOTES:

1. DESIGN BASIS: DEPROPANIZER COLUMN VER 1 OTS MODEL, DESIGN CASE. PRESSURES MPa g, MW 44.1 / 58.1 kg/kmol ASSUMED. FEED 60 m³/h = 31.8 t/h (634 kmol/h, 57 mol-% C3), -1 → 75 °C IN E-201 (67 % VAPOUR), TRAY 18 OF 27. PROPANE 32.1 m³/h = 16.1 t/h; BOTTOMS 27.8 m³/h = 15.6 t/h; VENT 72 Nm³/h. MASS CLOSURE +0.14 %.

2. TOP 1.70 / 45 °C, BOTTOM 109 °C, V-202 1.68, PUMP DISCH. P-201 2.36, P-202 2.26, E-201 3.50 MW (5.8 t/h LS), E-203 2.24 MW (3.8 t/h LS), E-202 3.96 MW (CW 330 m³/h, 15 → 25 °C). OVERHEAD 40.8 t/h, 8 in AT 8.5 m/s.

3. LINE SIZES PRELIMINARY FROM VELOCITY (LIQUID 1-2.5 m/s, VAPOUR ≤ 10 m/s) - TBC BY HYDRAULIC CALCULATION.

4. PRESSURE INSTRUMENTS C/W ROOT AND TEST VALVES; PUMP CASING VENTS/DRAINS; TYP, NOT SHOWN.

5. XV-201, XV-202, XV-209 FAIL CLOSED, OPERATED BY THE SIS (SIGNALS NOT SHOWN). HV-202, HV-209 ARE MANUAL BLOCKS.

6. FAIL POSITIONS: FV-201, TV-202, TV-208, LV-202, LV-203 FC; FV-204 (REFLUX) AND PV-210 (VENT) FO.

7. LT-202 / LS-202 AND LT-203 / LS-203 ON SEPARATE NOZZLES; PT-211 INDEPENDENT OF PT-210 (PIC-210).

8. P-201A MOTOR, P-201B STEAM TURBINE (DIVERSE DRIVE); P-202A/B MOTOR. SPARE AUTO-STARTS (SCHEDULE).

NOTES (CONT.):

8. TIC-227 (TRAY 27) CAN CASCADE THE FIC-204 SET POINT (CAS); NORMALLY MAN, FIC-204 ON ITS OWN SET POINT. SIGNAL NOT SHOWN.

9. PDI-212 COLUMN DIFFERENTIAL PRESSURE, TAPS ABOVE TRAY 27 AND IN THE SUMP VAPOUR SPACE (FLOODING INDICATOR).

10. PSV-201 4L6, SET 1.80 MPa g; CW FAILURE 40 852 kg/h (API L ORIFICE); FIRE CASE TBC; 5.9 % OPERATING MARGIN - HOLD.

11. V-201 SHOWN WITHOUT OVERPRESSURE PROTECTION IN THE OTS SCOPE - PSV REQUIRED, HOLD.

12. PSHH ON C-201 OVERHEAD TO TRIP REBOILER STEAM BEFORE PSV LIFT RECOMMENDED - HOLD (NOT IN OTS SCOPE).

13. DCS INDICATORS (TI, PI, FI, AI) IMPLY A FIELD TRANSMITTER; AI-219 C4+ IN PROPANE, AI-220 C3 IN BOTTOMS.

14. ALARM SET POINTS: SEE WORKBOOK (ALARM TABLE); TRIP SET POINTS: INTERLOCK SCHEDULE.

15. TRAYS NUMBERED FROM THE BOTTOM; ONLY FEED, CONTROL AND INDICATION TRAYS ARE LABELLED.

16. DESIGN CASE (REV A): NO V-201 LEVEL CONTROL, LT-201 LOCAL. REV B: LI-201 ON DCS, LAL/LAH-201, FAL-201 ON FIC-201. REV C: LIC-201 SETS THE FIC-201 SET POINT (CASCADE, RSP). BOTH ARE UPGRADES SELECTED IN THE OTS.

LEGEND

PROCESS LINE

UTILITY LINE (STEAM, COOLING WATER, RELIEF)

INSTRUMENT SIGNAL

MANUAL VALVE (GATE)

CONTROL VALVE, FO / FC = FAIL OPEN / CLOSED

SIS SHUT-OFF VALVE, SOLENOID, FAIL CLOSED

CHECK VALVE

FLOW ELEMENT (ORIFICE)

STEAM TRAP

RELIEF VALVE (PSV)

FIELD-MOUNTED INSTRUMENT

DCS FUNCTION (OPERATOR CONSOLE)

SIS FUNCTION

SERVICE CODES: P PROCESS HYDROCARBON · LS LP STEAM
CW COOLING WATER · DR DRAIN · COND. STEAM CONDENSATE
LINE NO. = SIZE - SERVICE - NUMBER (SIZES TBC, NOTE 2)

INTERLOCK SCHEDULE · SIS

TAG	CAUSE	SETTING	DELAY
LSLL-202	C-201 SUMP LOW-LOW *	< 5 %	5 s
LSHH-203	V-202 LEVEL HIGH-HIGH	> 92 %	5 s
HS-ESD	ESD PUSHBUTTON (DCS)	MANUAL	0 s
FSSL-201	LOW FEED FLOW **	< 15 m³/h	5 s
PDSL-230	LOW P-202 DIFF. PRESS.	< 150 kPa	3 s
PSV-201	C-201 OVERPRESSURE	1.80 MPa g	MECH.

EFFECTS:

LSLL-202: CLOSE XV-209 & XV-202; TIC-208, TIC-202 MAN 0 %.

LSHH-203 / HS-ESD: AS LSLL-202, PLUS CLOSE XV-201, FIC-201 MAN 0 % (FEED STOPPED).

FSSL-201: AUTO-START P-201B (NOT A TRIP).

PDSL-230: AUTO-START P-202B (NOT A TRIP).

* ARMED ONLY WHEN SIS IS IN RUN (HEAT ON).

** ONLY WITH FV-201 > 20 % AND XV-201 OPEN.

RESET BLOCKED WHILE A CAUSE IS PRESENT. RE-OPEN XV-209 / XV-202 NEEDS: FI-216 ≥ 250 m³/h, LIC-202 ≥ 30 %, TIC-208 & TIC-202 MAN ≤ 10 %, HV-209 & HV-202 OPEN.

REV	DESCRIPTION	DATE
B	LI-201 ON DCS, LAL/LAH-201, FAL-201 (OPTION)	--
C	LIC-201 CASCADE; OTS SCREEN DWG (USED IN DECKS)	--
D	ALTERNATIVE VERSION, EPC STANDARD	30-SEP-26
DEPROPANIZER COLUMN VER 1 · OTS		
P&ID — C-201 · ALTERNATIVE VERSION		
DRAWN: APINAN SOOTTITANTAWAT	CHK'D: --	APP'D: --
DWG NO. DC3-V1-PID-001	REV D	SCALE N.T.S.
		DATE: 30-SEP-26
		SHEET 1 OF 1